Housing Decision Analysis

Option A: Trailer → Permanent Home vs Option B: Buy Now → Rent → Buy Again

Muscle Shoals, AL | 10-Year Horizon | 1,000 Monte Carlo Paths | Generated April 10, 2026

Simulation Parameters

Parameter	Option A	Option B
Simulation length	120 months (10 yrs)	← same
Monte Carlo runs	1,000	← same
Monthly income	$5,100	← same
Starting cash	$20,000	← same
Phase 1 upfront cost	$4,000 (fixup)	$8,373 (buy house)
Phase 1 monthly gas	$600/mo	$300/mo
Phase 1 ends	Month 25	Month 38
Phase 2 home price	$250,000	$250,000
Phase 2 mortgage	$1496.93/mo	$1496.93/mo
Phase 2 upfront	$37,500	$37,500
Rental income (B only)	None	$700–$1000/mo from Mo 38
Vacancy rate (B only)	N/A	10%
Passive income goal	N/A	$300/mo
Home appreciation	3.0%/yr	← same
Expense inflation	3.0%/yr	← same
HYSA return	4.5%/yr	← same
Time value of commute	$560/mo	← same

Affordability Check

The following table shows whether each scheduled purchase is actually affordable based on projected cash at the time of purchase. If a purchase is not affordable at the scheduled month, the simulation automatically delays it until cash is sufficient.

Purchase	Scheduled Month	Cash Available	Cost Required	Left Over	Status
Option A — Trailer fixup	Month 1	$20,000	$4,000	$16,000	✓ **Affordable**
Option A — Permanent home	Month 25	$87,840	$37,500	$50,340	✓ **Affordable**
Option B — Buy House 1	Month 1	$20,000	$8,373	$11,627	✓ **Affordable**
Option B — Buy House 2	Month 38	$93,332	$37,500	$55,832	✓ **Affordable**

Purchase Delay Statistics (across all 1,000 Monte Carlo paths):

	Scheduled Month	Median Actual Month	% of Paths Delayed
Option A home purchase	Month 25	Month 25	0%
Option B House 2 purchase	Month 38	Month 38	0%

Key Results Summary

Metric	Option A	Option B	Winner
Mean net worth — Month 120 (no time value)	$282,909	$336,468	**B**
Mean net worth — Month 120 (with time value)	$198,232	$336,319	**B**
10th pct net worth (no TV)	$281,028	$331,532	**B**
90th pct net worth (no TV)	$284,677	$341,045	**B**
Break-even month (median, no TV)	—	Month 25	—
% paths B wins (no TV)	—	100%	—
Break-even month (median, with TV)	—	Month 6	—
% paths B wins (with TV)	—	100%	—
% paths hit $300/mo passive income	N/A	55%	—
Median month goal achieved	N/A	Month 100	—
House 1 equity at final month (B)	—	$50,810	**B**
House 2 equity at final month (B)	—	$101,331	**B**
Total rental income earned (mean)	—	$70,405	**B**
Total upfront costs (lifecycle)	$41,500	$45,873	—

Milestone Snapshots

Milestone	A Net Worth	B Net Worth	B Total Equity	B Passive Income
Year 5 (Mo 60)	$150,224	$158,613	$68,813	$70/mo
Year 10 (Mo 120)	$282,909	$336,468	$152,141	$166/mo

Chart 1: Net Worth — 10-year horizon. Option A (trailer→home) vs Option B (buy→rent→buy). Shaded bands show P10–P90 Monte Carlo uncertainty. Phase transition lines shown.

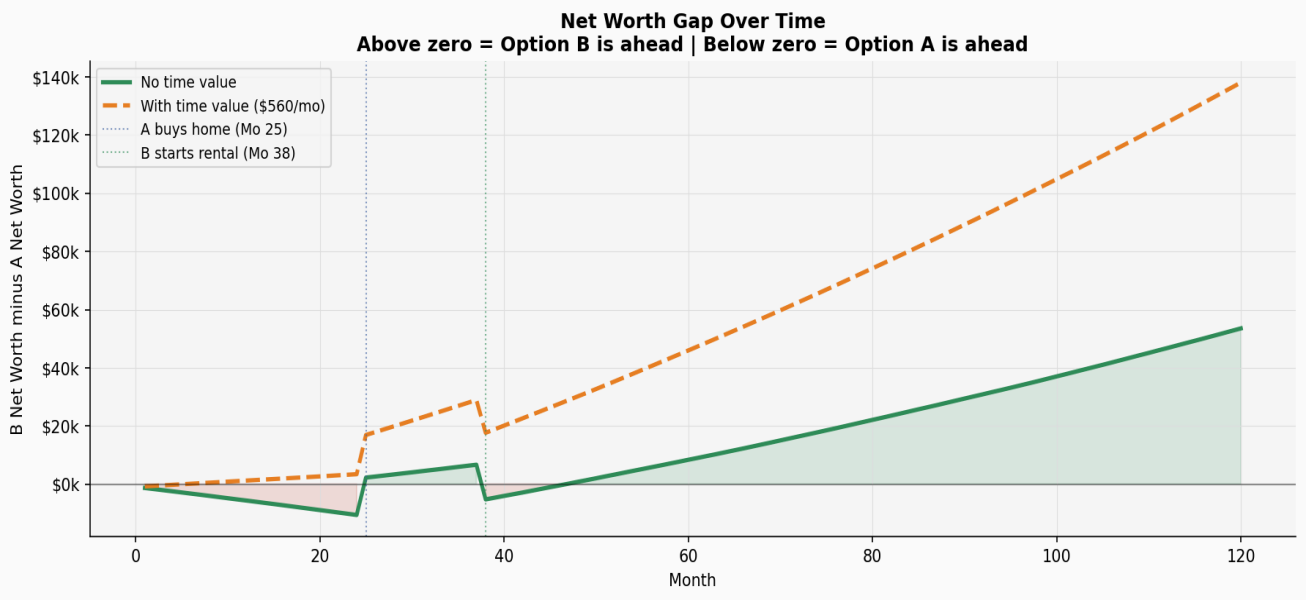

Chart 2: Net Worth Gap (B minus A). Above zero = Option B is ahead. Shows when and by how much each option leads.

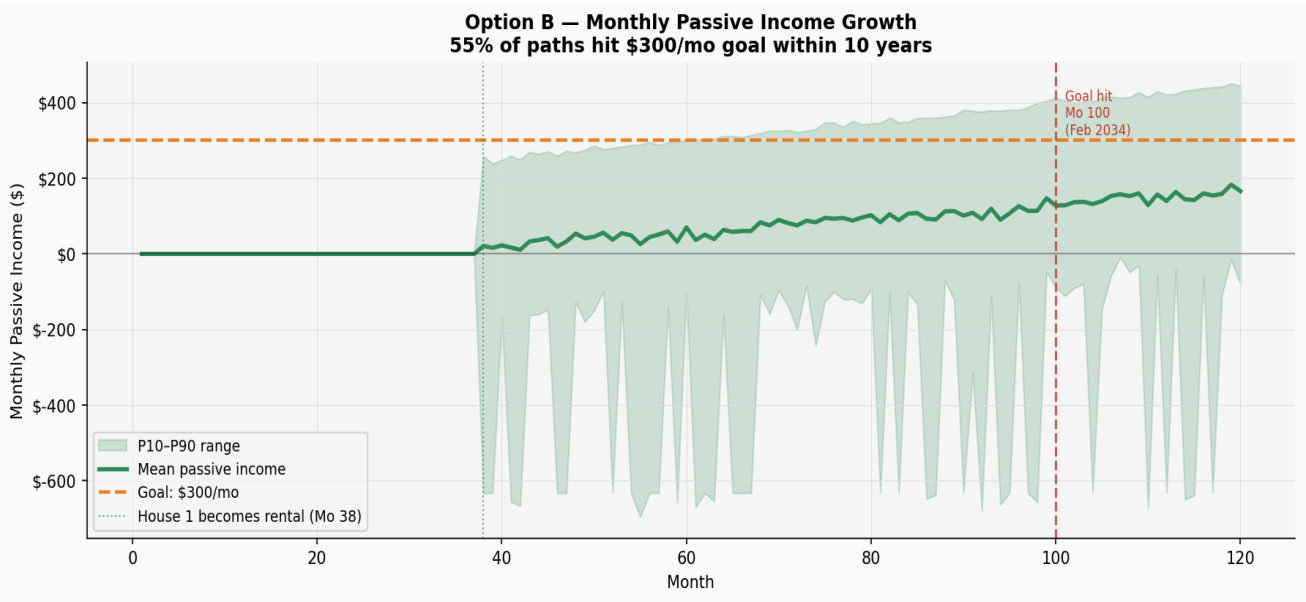

Chart 3: Option B monthly passive income growth. Goal line at $300/mo. 55% of simulation paths achieve this within 10 years.

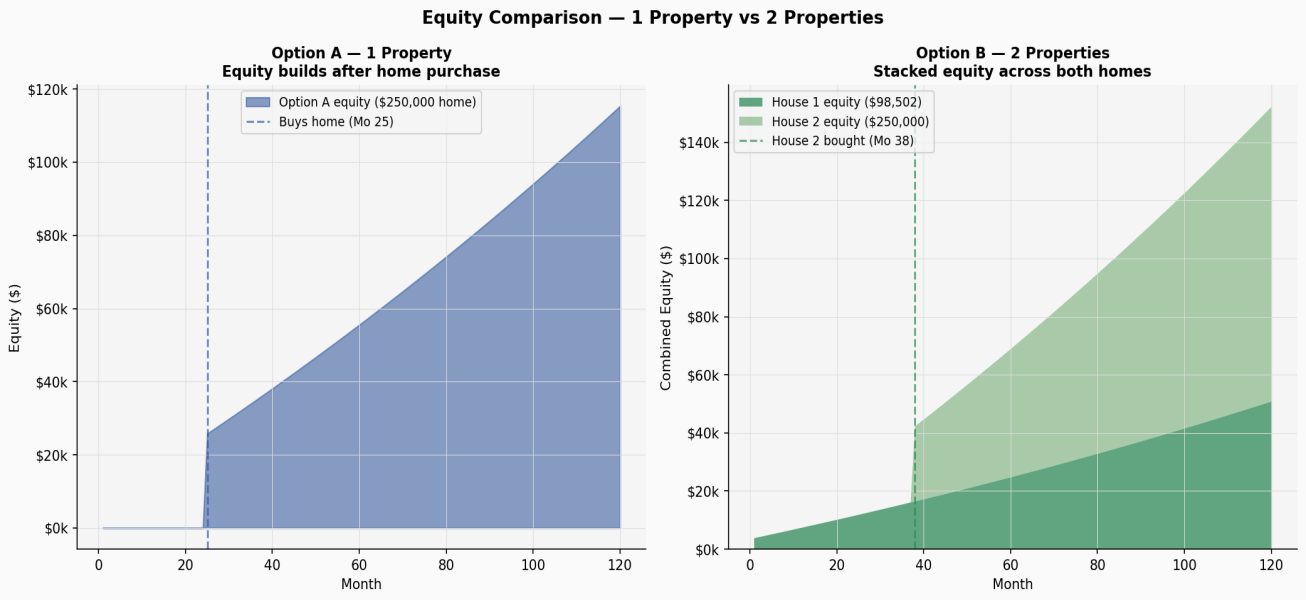

Chart 4: Equity comparison. Option A builds equity in one home; Option B builds stacked equity across two properties.

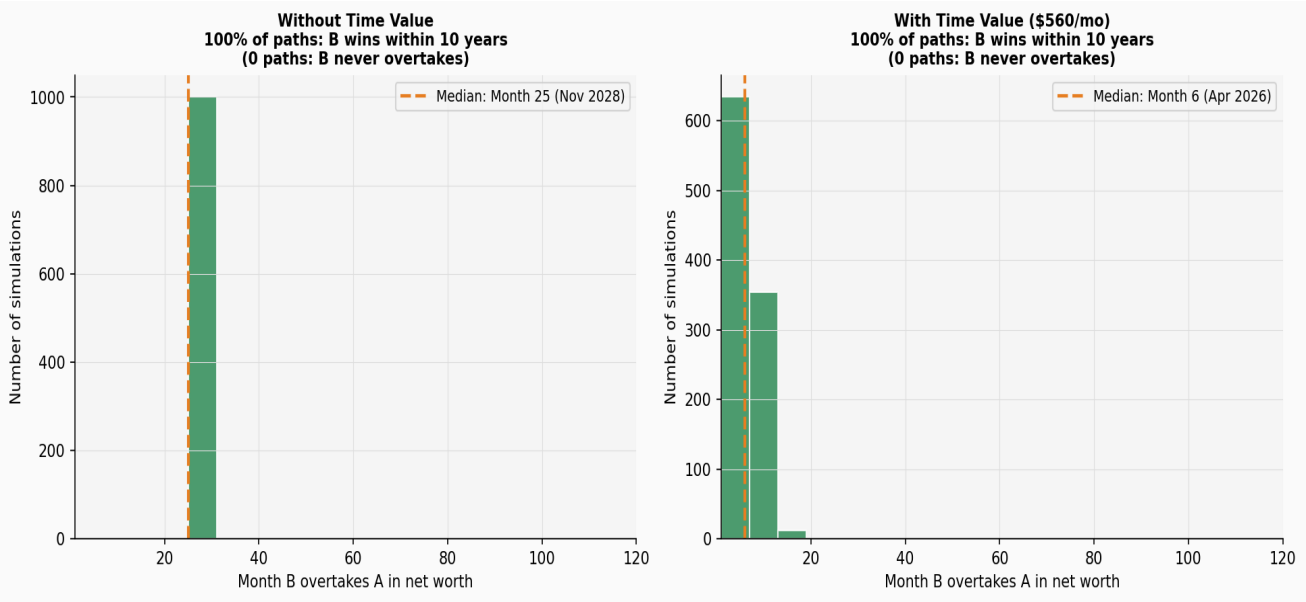

Chart 5: Break-even distribution — month B net worth first exceeds A for 3+ months.

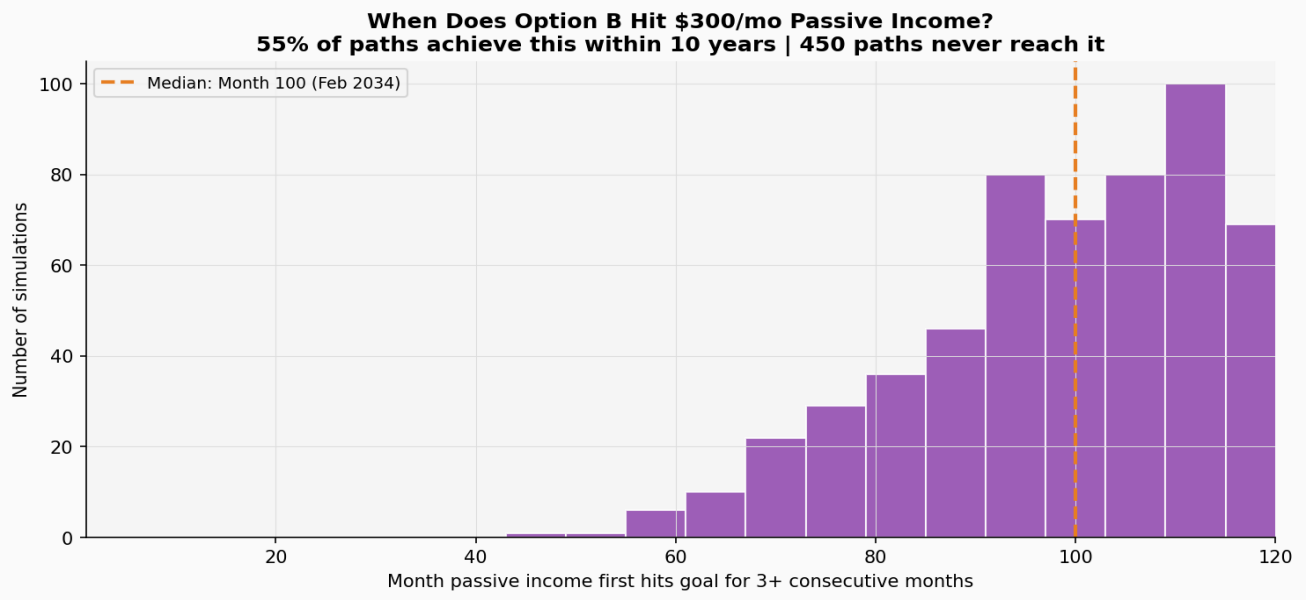

Chart 6: Goal achievement distribution — month Option B first sustains $300/mo passive income.

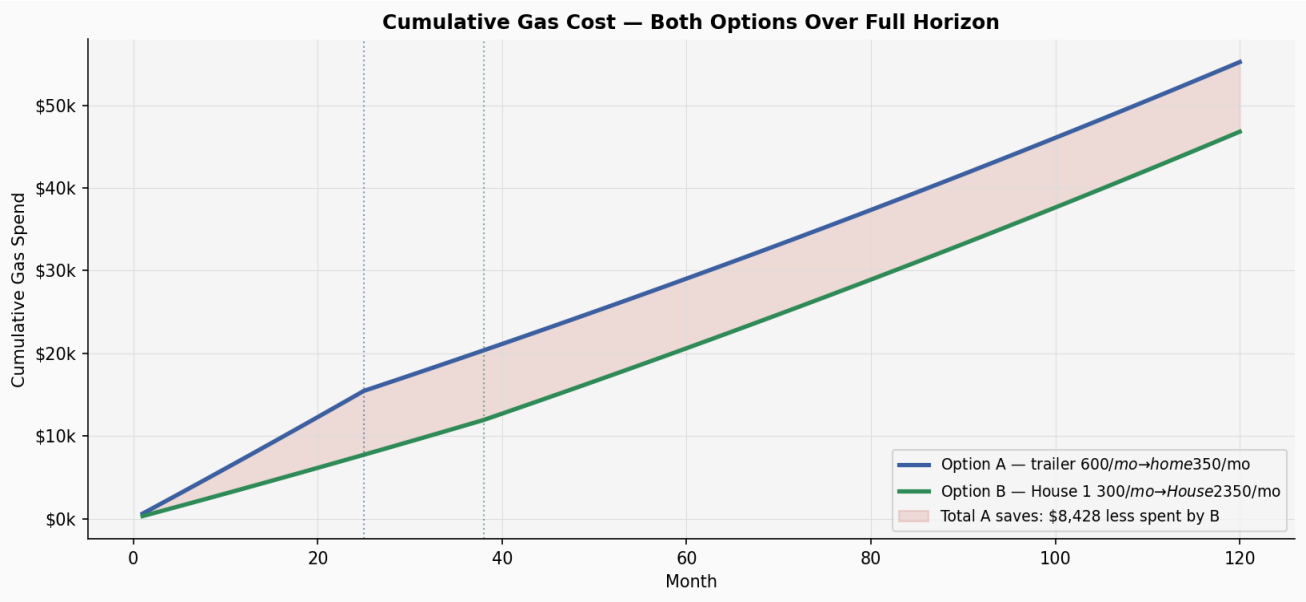

Chart 7: Cumulative gas costs. Both options transition gas rates at their respective move-out months.

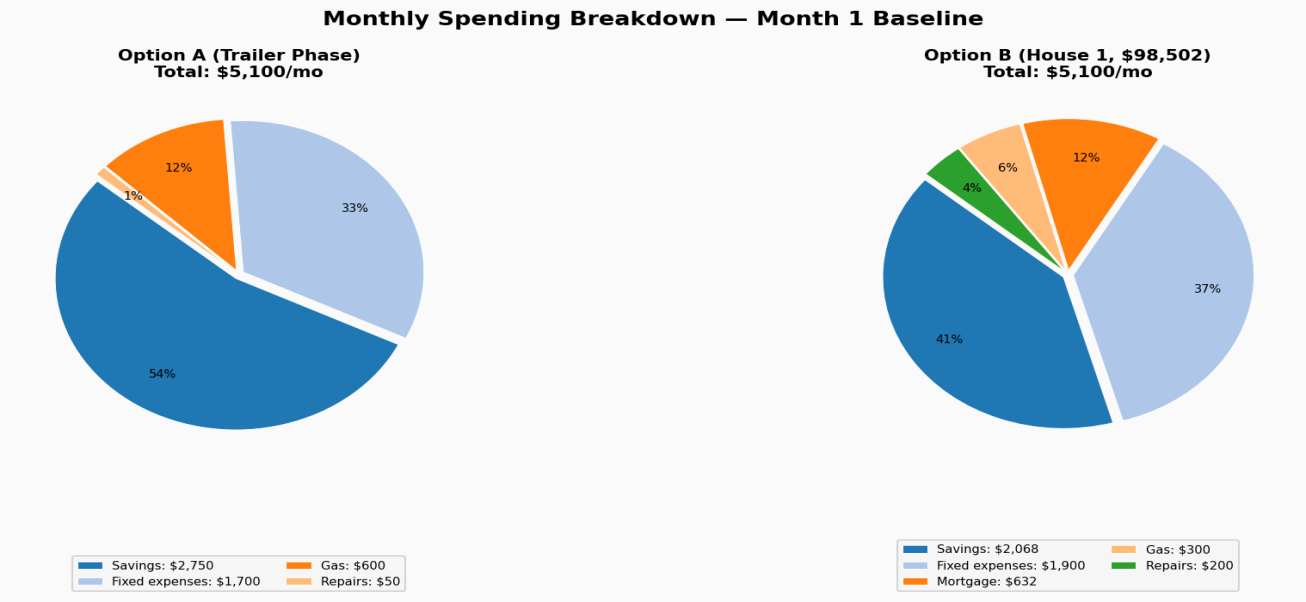

Chart 8: Monthly spending breakdown at Month 1 baseline using actual config values.

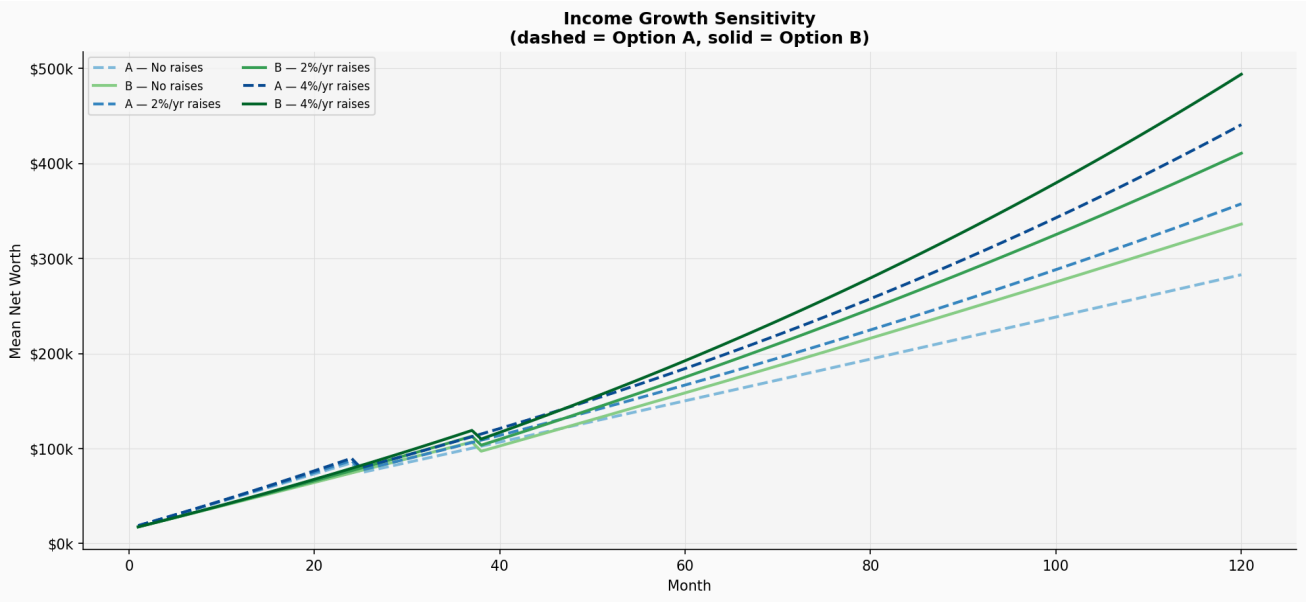

Chart 9: Income growth sensitivity — net worth across three raise scenarios.

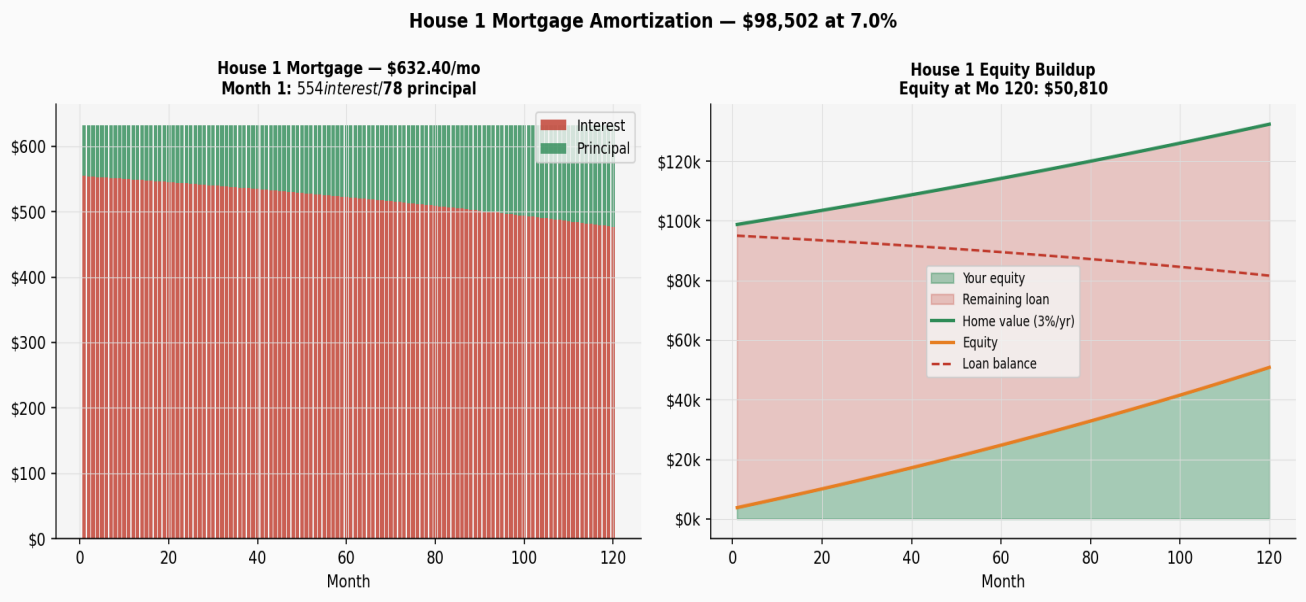

Chart 10: House 1 mortgage amortization. $98,502 at 7.0% — interest/principal split and equity growth.

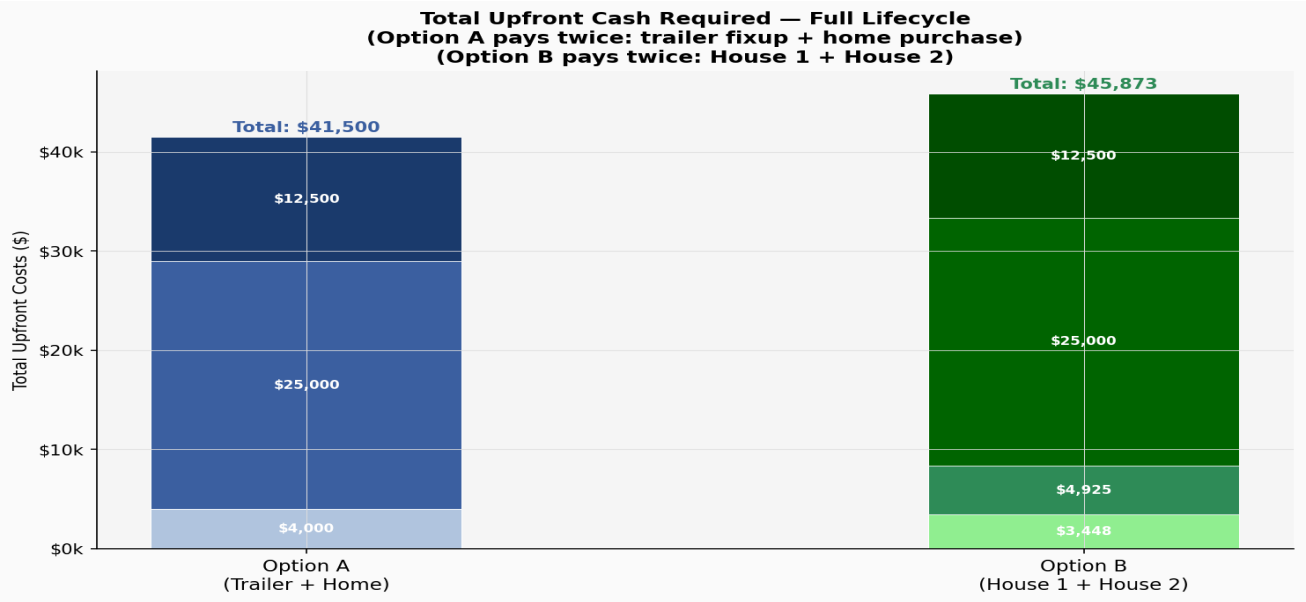

Chart 11: Total lifecycle upfront costs. Both options pay twice — Option A pays for trailer fixup then permanent home; Option B pays for House 1 then House 2.

Analyst Verdict

At 10 years, Option B leads by $53,559 in the base scenario (no time value). With $560/mo commute time value included, Option B leads by $138,087.

Option B achieves the $300/mo passive income goal in 55% of simulation paths, with a median achievement month of 100 (Feb 2034).

Key risks to monitor:

- Option B cash after House 1 closing: $11,627. Repair spikes in year 1–2 are the primary short-term risk.
- Option A makes two separate purchases totaling $41,500 in upfront costs. Option B also pays twice but the second payment (House 2) is partially funded by rental income.
- Vacancy risk on House 1 rental modeled at 10%. Good tenant screening is worth the time.
- If income drops significantly, Option A's lower early fixed costs (trailer phase) provide more breathing room.